



Samsung's LED Flames

Samsung's new induction cookers will now have LED lights to tell you when the heating is on <http://dgit.in/LEDFlames>

A heart is reborn

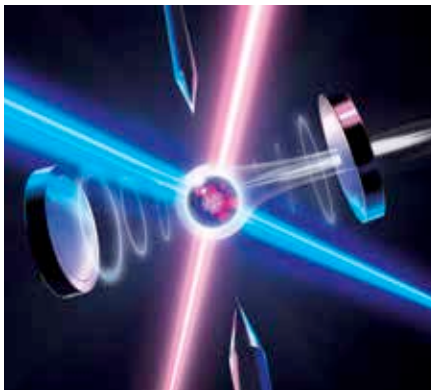
Australian scientists successfully managed to transplant a dead heart into a patient. Find out how at: <http://dgit.in/1ts6yqS>

Tomorrow's tech

helping impaired people communicate to the ability to reach out to soldiers in war scenarios - the choice is ours.

Unhackable Quantum Internet

With the interconnectedness of the Internet and the sensitivity of personal data online researchers have been working away at using quantum technology to secure the Internet. The technology in question is known as quantum key distribution or QKD.



Photons can hold any one of four positions which can be sequenced for a key

The method of using quantum keys to encrypt messages is being experimented across the world in places such as China and Switzerland. Using this technique, Internet networks transmit photons in specific quantum states to create a cryptographic key. Using quantum key distribution, data can be encrypted over ordinary, non-quantum networks due to its higher degree of mathematical complexity.

This means that in order to decrypt the message a very powerful computer is necessary. But the beauty of the QKD protocol is that if anyone was to attempt to intercept a quantum key, it would alter the quantum states of the photons and alert users that the key is under attack.

With successful small scale networks in the United States, China and Switzerland proving unbeatable efforts are being made at securing any loopholes such as hardware hacks and social hacks to further reinforce the system. At the present moment, the network range is restricted due to the issues of quantum stability over long distances, but that problem

is being addressed with the development of quantum repeaters that would use quantum entanglement to create a globally secure network. The research is underway and we can expect it to be the next iteration of the Internet of today.

Cheap Nanotube Hydrogen Fuel

The world is in an energy crunch. With fluctuating fuel prices, diminishing supplies of fossil fuels, nascent sources of renewable energy and the impact of climate change making the global situation dire. To make matters worse, the need to feed, house and support the seven billion and rising population of the planet makes the discovery of a viable energy alternative a must.

Under these circumstances the invention of the technology that allows the cheap manufacturing of clean burning hydrogen fuel is momentous. Researchers at Rutgers University, New Jersey, have developed a method of using carbon nanotubes to manufacture commercially useful hydrogen fuel from water. The usual method of converting water to hydrogen fuel is extremely expensive as it uses platinum for its electrolysis reactions.

But with the development of the new sustainable chemical catalyst in the form of "noble metal-free nitrogen rich carbon nanotubes". The bottom line being that with the use of these new carbon nanotube catalysts in electrolysis reactions researchers can set up water to hydrogen fuel conversion systems in a variety of conditions. By simply splitting the water molecules to create clean, commercially viable hydrogen fuels, the system can be adapted to a number of settings and environments, as well as be coupled with existing energy producing systems like solar, hydro, wind and nuclear energy.

This discovery directs the use of nanomaterials towards one of the most important problems of our generations and has the potential to change the way the world fuels itself in the future.

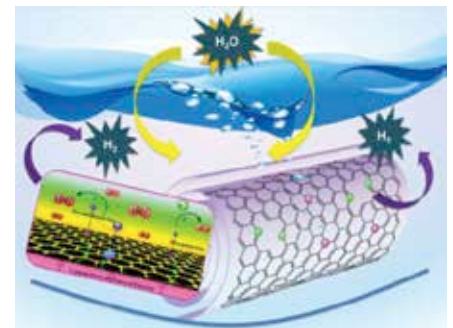
Liquid Data Storage Devices

We are surrounded by data and most of it's smashable. It can be anything from a cell phone to a cloud server - in the end all

the data of the digital world exists in some solid form ready for smashing. But with the recent developments in nanoparticle studies at the University of Michigan and New York University, we may soon find data dripping from every surface. Researchers at these universities have designed a method for storing digital data in liquid form and that too at immensely high densities.

A normal one terabyte hard-disk would come in at the size of a small cell phone (47.5 milliliters of volume from Western Digital) while a one terabyte liquid storage would fit in a tablespoon (which is about 15 milliliters of volume). The usage of "wet" or "soft" storage mediums as been an ongoing research objective for data storage and researchers have finally found a working system. By using suspended clusters of nanoparticles in a liquid they have discovered that they can store more data than the typical binary systems computer.

Since these nanoparticles are in the form of one central sphere attached to up to 12 particles in different configurations, the capacity of data storage increases by



The carbon nanotubes act as a catalyst for the electrolysis reaction separating water molecules.

an exponential factor. A cluster consisting of 12 particles attached to a single sphere can create nearly 8 million unique configurations as compared to the binary format of 1 and 0. This equals 2.86 bytes of data storage per cluster. To put that in context - the tablespoon of storage we mentioned earlier only uses a three percent concentration of these 12 particles in the liquid to attain a terabyte of storage.

Scientists are now working on inventing a fast and reliable method of reading